



Data Analytics Internship

Two Projects · End-to-End · From Raw Data to Insights

</> PYTHON

MySQL

POWER BI

MACHINE LEARNING

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VegaIT Internship 2026

Two Projects, One Goal

From raw data collection to a fully functional analytics platform, both projects follow the same end-to-end process. The data is extracted, cleaned, stored in a Star Schema data warehouse, and visualized in Power BI. What makes them different is the domain, the data source, and the complexity of the dataset.

PROJECT 1

SmartStore Analytics

Retail sales intelligence platform

- ETL Pipeline & Data Cleaning
- Star Schema Data Warehouse
- Power BI Dashboard (6 pages)
- ML: Order Profit Prediction

PROJECT 2

Euroleague Analytics

European basketball data platform

- Official API Integration
- ETL & Star Schema (3 fact tables)
- Power BI Report (4 pages)

Projects Tech Stack

SmartStore

- Python 3.13 · Pandas
- SQLAlchemy · MySQL
- Power BI · DAX
- scikit-learn · XGBoost · scipy

Euroleague

- Python 3.13
- euroleague_api · MySQL
- Power BI · DAX

Shared

- Technologies**
- Data Warehouse · ETL Pipeline
 - Data Warehouse · Git · GitHub

 PYTHON MYSQL POWER BI GITHUB

ANALYTICS

From 6 Raw CSV Files to a Complete Analytics Platform

2,851

Total Orders

1,130

Unique Customers

\$3.10M

Total Sales

\$461K

Total Profit

6

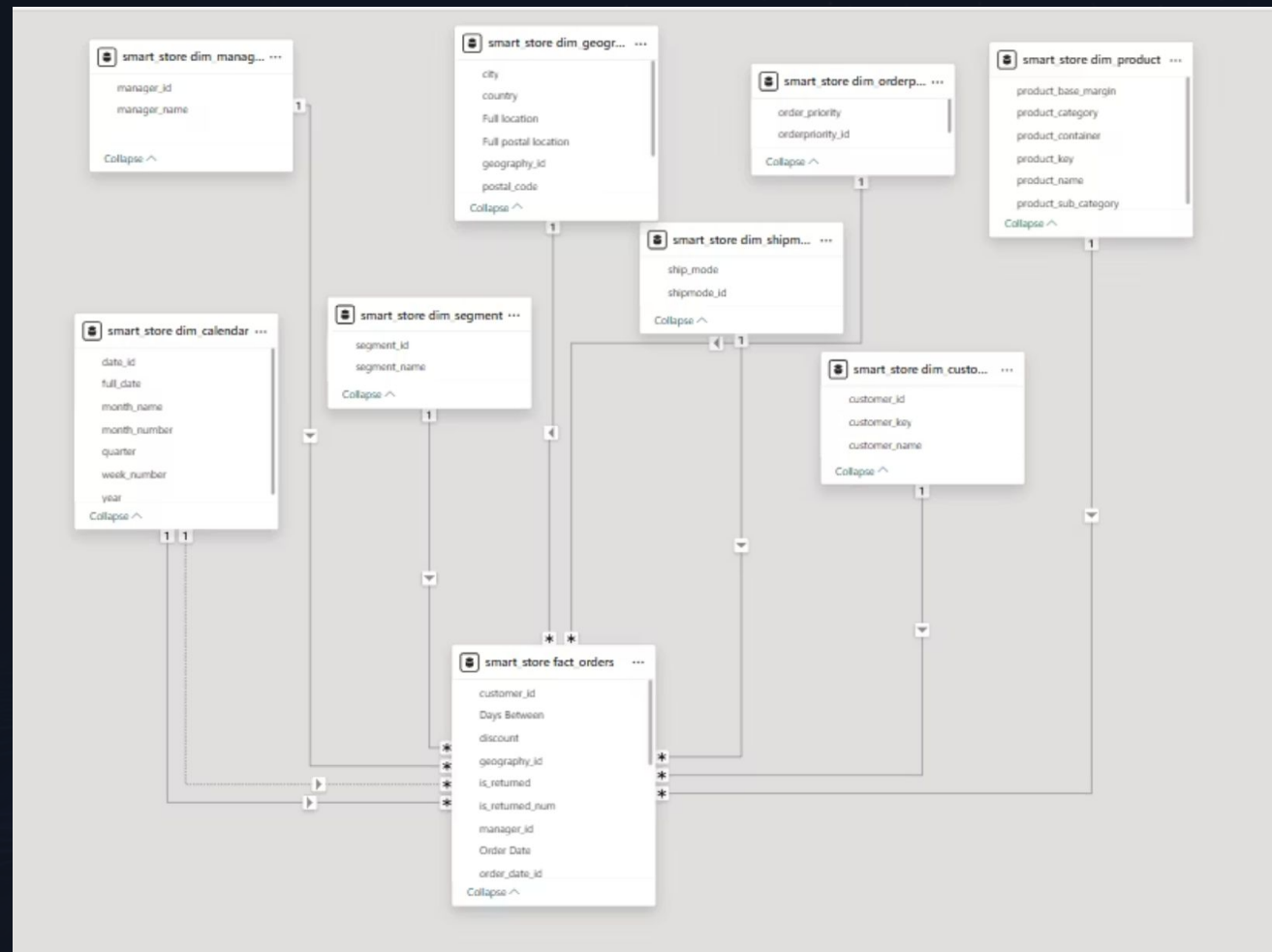
Power BI Pages

4

US Regions

📄 1,951 original orders + 900 via Incremental Load automation

Star Schema + ETL Pipeline



Data Quality — Problems Solved

Trailing spaces → `TRIM()`

Postal codes → `LPAD()` to 5 digits

16 nulls → replaced with median

2 duplicate Row IDs → removed

MySQL Boolean → Power Query fix

📄 **Schema decision:** Started with Snowflake → switched to Star Schema for Power BI filter propagation and simpler DAX.

Dashboard + Machine Learning

Power BI — 6 Pages

1. Overview — KPI cards, dynamic date slicer
2. Table — full detail, 4 slicers
3. Analysis of Sales — Decomposition Tree
4. Month over Month — Current vs Previous
5. Customers and Managers analytics — Bookmark toggles
6. H1 vs H2 2015

ML Results — Order Profit Prediction

Model	MAE	R ² Score
Random Forest ✓	\$266	0.271
XGBoost v1	\$281	0.096
XGBoost v2	\$279	0.083

- ❏ Random Forest won — small dataset (2,851 rows) favors ensemble methods over gradient boosting. Kruskal-Wallis $p=0.428$ — no significant regional difference.

Examples:

```

• USE smart_store;
• SELECT
    p.product_sub_category,
    SUM(f.sales) AS Total_Sales,
    RANK() OVER (ORDER BY SUM(f.sales) DESC) AS Sales_Rank
FROM fact_orders f
JOIN dim_product p ON p.product_key = f.product_id
GROUP BY p.product_sub_category
ORDER BY Sales_Rank
LIMIT 5;
    
```

```

• SELECT
    m.manager_name,
    SUM(f.profit) AS Total_Profit,
    RANK() OVER (ORDER BY SUM(f.profit) DESC) AS Profit_Rank
FROM fact_orders f
JOIN dim_manager m ON m.manager_id = f.manager_id
GROUP BY m.manager_name
ORDER BY Profit_Rank;
    
```

MySQL query

Results:

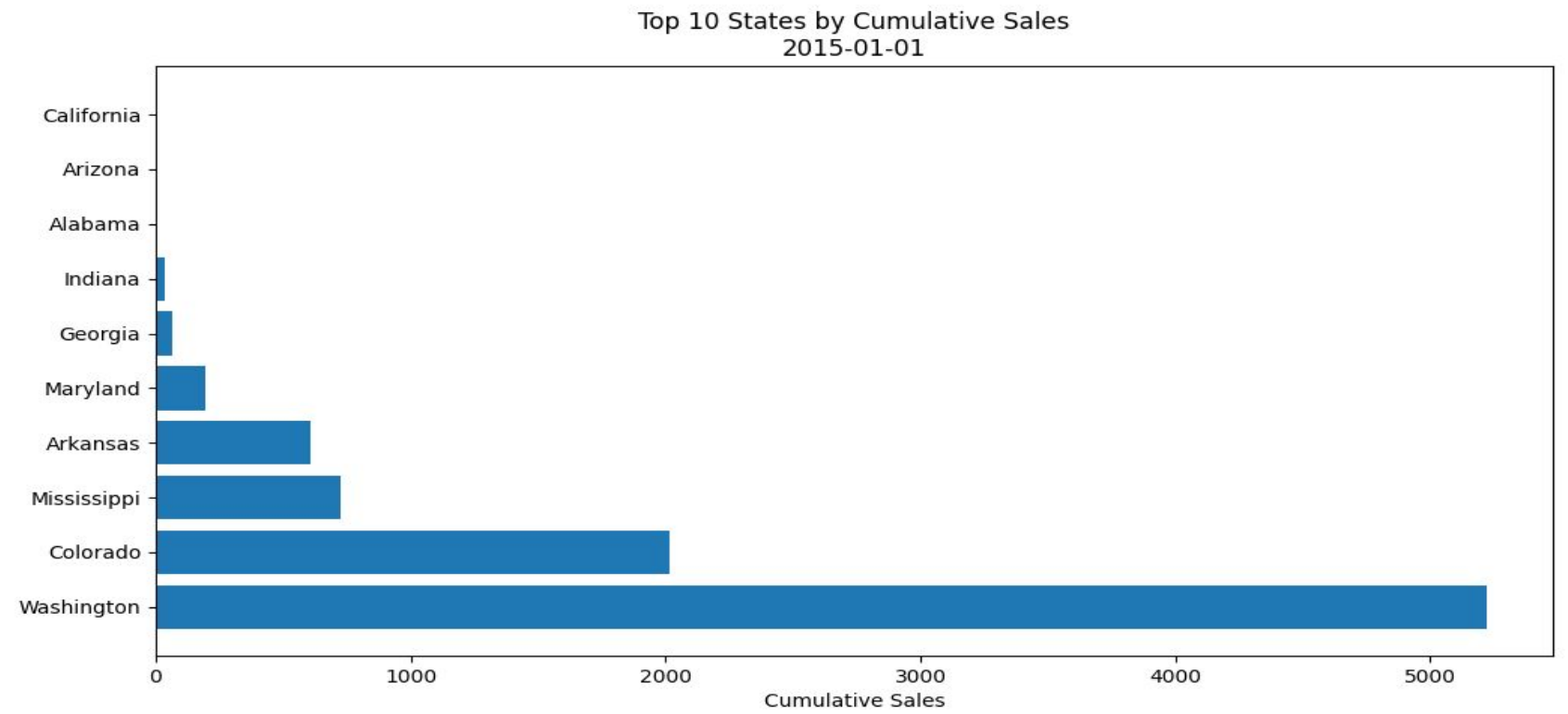
	product_sub_category	Total_Sales	Sales_Rank
▶	Office Machines	362767.47	1
	Binders and Binder Accessories	317069.76	2
	Telephones and Communication	316211.48	3
	Chairs & Chairmats	298729.89	4
	Storage & Organization	270778.37	5

This query joins the fact table with the product dimension, calculates total sales per sub-category, and ranks them using a window function. LIMIT 5 returns only the top five.

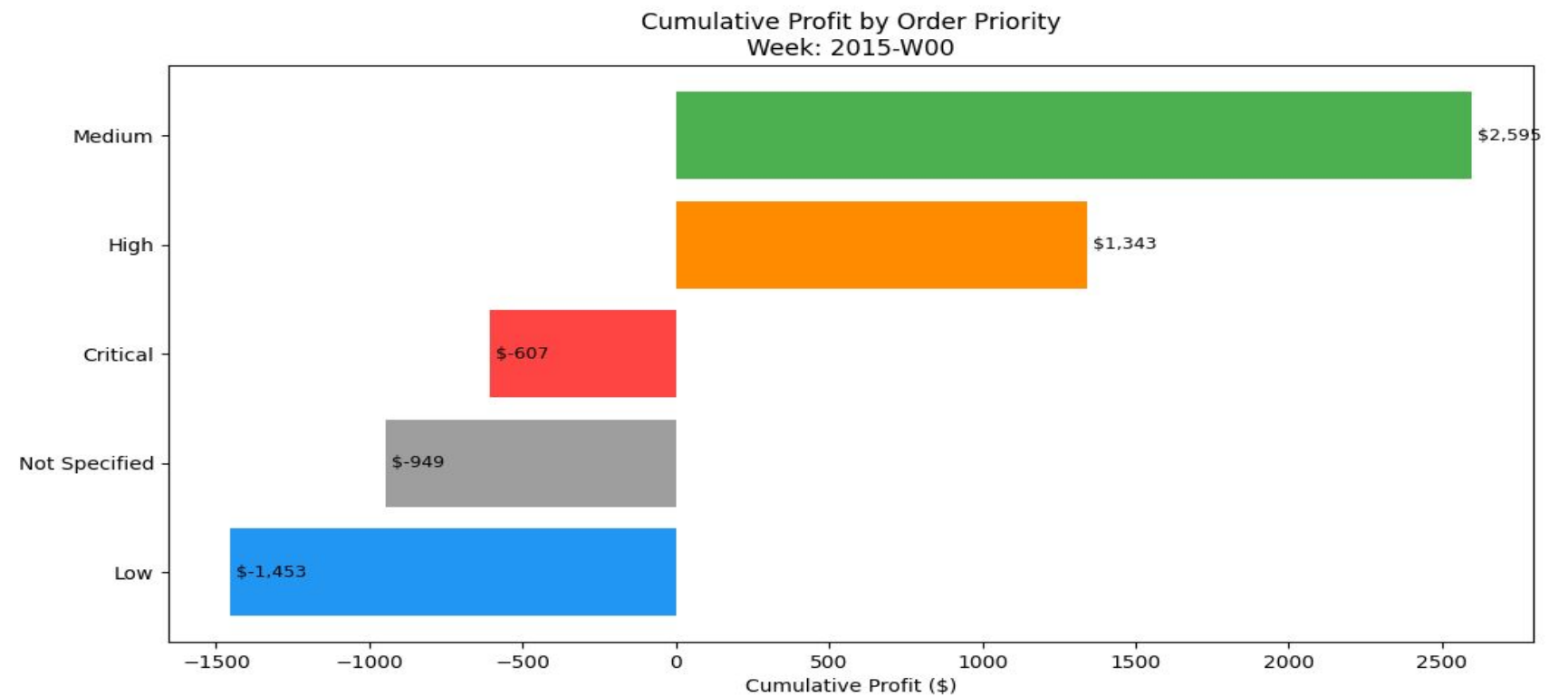
	manager_name	Total_Profit	Profit_Rank
▶	Chris	153406.65	1
	Erin	143487.37	2
	William	132903.11	3
	Sam	32173.75	4

This query joins the fact table with the manager dimension and ranks all managers by total profit.

TOP 10 States by Cumulative Sales



Cumulative Profit by Order Priority



10 Seasons of European Basketball — From API to PowerBI

10

Seasons

2015–2024

2,897

Games Analyzed

752

Unique Players

1,989

Player Season Records

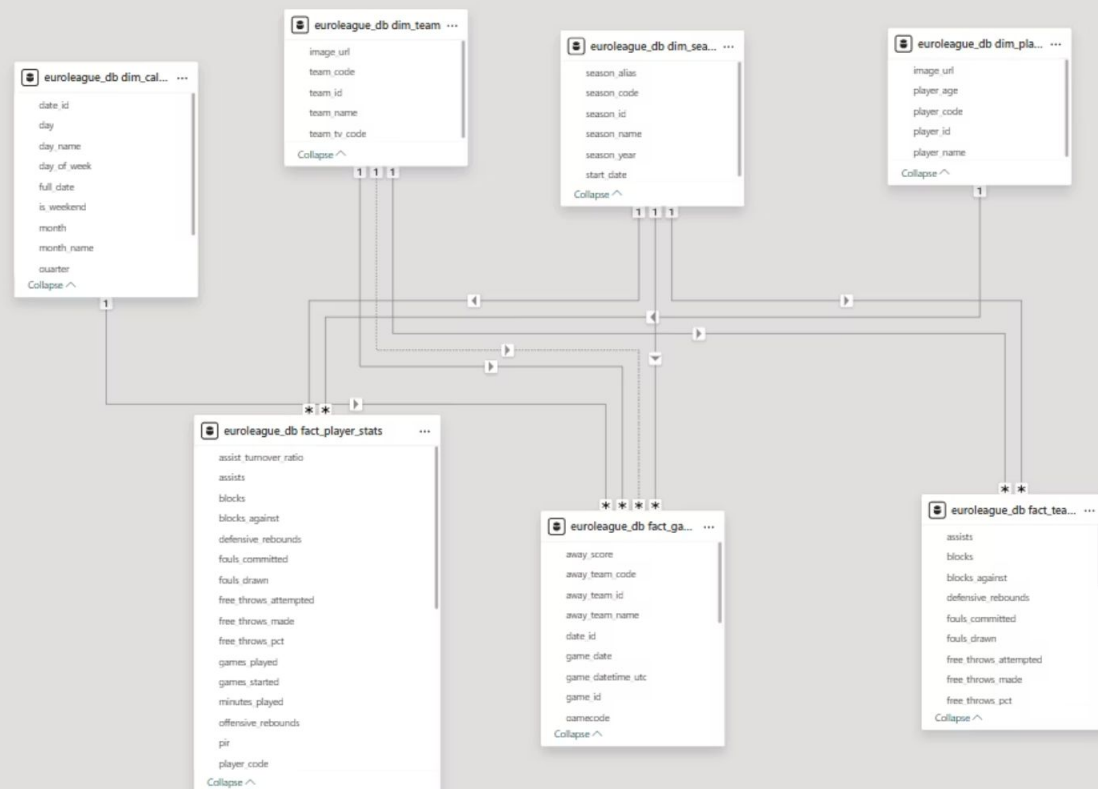
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Power BI Pages

 Data source: Official Euroleague API

Star Schema — 3 Fact

Tables



Key ETL Fix

`get_range_seasons()` returned `season_code=ALL` → only 205 rows. Fixed with manual loop per season → **1,989 rows** (10× more data).

- ❏ **Role-Playing Dimension:** `fact_games` has two relationships to `dim_team` — `home_team_id` (Active) and `away_team_id` (Inactive via `USERELATIONSHIP` in DAX).

Power BI Report — Key

Insights

Dashboard Pages

Overview — KPI cards + scoring trend

Team Performance — Bookmarks + Top N parameter

Player Performance — Field parameter (PIR / Points / Assists / Rebounds)

Season Analysis — Phase parameter + COVID impact

Key Findings

Avg score: **77 pts (2015) → 84 pts (2024)**

Real Madrid: most dominant team last 10 years

Home win rate: **62.4%** → dropped to **55%** in COVID season

Best PIR(Performance Index Rating): **Shane Larkin · 25.8**
(2018–19)

Playoff games score **5.7 pts more** than Regular Season



COVID (2020–21): No fans = no home advantage.
Natural experiment — **7% drop** in home win rate.

Thank you for your attention!